

Dan Nguyen

223-255-1313 | sidan.nguyen@gmail.com | [linkedin.com/in/dannguyen111](https://www.linkedin.com/in/dannguyen111) | dannguyen111.github.io | github.com/dannguyen111

EDUCATION

Gettysburg College

Gettysburg, PA

B.S. in Computer Science and Mathematics (Rank: 7/547) | GPA: 4.0

Jan. 2023 – May 2026

- Honors: Phi Beta Kappa, Honors in Mathematics, Earl E. Ziegler Mathematics Award, Pi Mu Epsilon

EXPERIENCE

AI Engineer Intern

June 2025 – Aug. 2025

KPMG

Berlin, DE

- Re-recruited based on Summer 2024 performance; designed and owned an end-to-end data pipeline system serving KPMG's Global Audit division across multiple international teams.
- Engineered an **ETL pipeline** in **Python** to parse and transform unstructured PDFs into structured **Neo4j** knowledge graphs, improving downstream QA processing efficiency by **30%**.
- Refactored a multi-stage preprocessing codebase into five modular **Python** pipeline components, improving maintainability and enabling reproducible fine-tuning workflows for the team.

Data Scientist Intern

May 2024 – Aug. 2024

KPMG

Berlin, DE

- Developed a deterministic rule-based baseline and a **stochastic unsupervised algorithm** for peer group clustering across all public banks, achieving a **15% improvement** in cohesion and separation.
- Designed quality metrics, benchmarked methodologies, and delivered the selected model for integration into production risk assessment tooling.

PROJECTS

Palamedes: Fraternity Management Platform | Django, PostgreSQL, AWS, Stripe

Dec. 2025 – Present

- Architected a full-stack, multi-tenant web application in **Django** and **PostgreSQL** with **AWS S3**; leveraged **Claude Code** to accelerate RBAC scaffolding and automate boilerplate across the multi-tenant data model.
- Engineered a custom **Role-Based Access Control (RBAC)** system with hierarchical boolean permissions, dynamically routing approval workflows based on a user's elected executive position.
- Integrated the **Stripe API** with asynchronous webhook listeners to automatically reconcile member balances on payment confirmation, eliminating manual treasurer reconciliation.

APPC Archive Assistant | Django, React, PostgreSQL (pgvector), Docker | Live Site

Jan. 2026 – May 2026

- Engineered a full-stack **RAG** application with a **Django** REST API, **React** frontend, and **pgvector** embeddings in **PostgreSQL**; containerized with **Docker** and deployed using **Claude Code**
- Implemented a **self-querying retriever** using **LlamaIndex** and **Pydantic** to extract structured metadata filters from natural language queries; architected a **multi-model fallback chain** (GPT-4o-mini, Gemini 2.5 Flash) with an admin cost-tracking dashboard.
- Built interactive **source citation** UI linking AI responses directly to source PDFs, ensuring transparency and explainability for end users.

TwinMind Copilot | Django, React, Docker | Demo | GitHub

May 2026

- Designed a split-deployment architecture: **React/Vite** frontend on Vercel and a containerized **Django** REST API on Fly.io, enforcing strict **CORS** policies across environments.
- Built a real-time audio pipeline batching microphone input for low-latency transcription, feeding a live inference engine that classifies and surfaces actionable meeting insights.
- Used **Claude Code** to rapidly prototype the token-budget manager and settings system, reducing development time on context-management logic.

Saint Petersburg AI Player | Java, Python | GitHub

Aug. 2025 – Dec. 2025

- Implemented a **Monte-Carlo Tree Search** agent with UCT selection, random rollout simulation, and **ML-trained evaluation functions** (logistic regression, random forest), achieving **1st place** in a class tournament.
- Ran large-scale simulations to surface statistically significant card imbalances, producing a quantitative balance report with cost-adjustment recommendations.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML/CSS, SQL

Frameworks & Libraries: Django, React, Bootstrap, LlamaIndex, scikit-learn, pandas, NumPy

Developer Tools: Git, Docker, AWS, Azure DevOps, PostgreSQL, pgvector, Neo4j, Stripe API, Claude Code, Codex